

GEOMETRIC QUANTIZATION BY PATHS

PART II: CONSTRUCTION OF THE PREQUANTUM GROUPOIDS

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ABSTRACT. In Part I, we constructed the prequantum groupoid for connected and simply connected parasymplectic spaces. This second part extends the construction to *arbitrary connected parasymplectic diffeological spaces* (X, ω) . The presence of a non-trivial fundamental group $\pi_1(X)$ introduces an obstruction to the additivity of the naive integration function on the space of loops, measured by a surfacic 2-cocycle.

We resolve this obstruction by disarticulating the periods into a canonical three-tiered geometric hierarchy: spherical periods, toric periods, and surfacic periods. The *Total Group of Periods* $P_\omega \subset \mathbf{R}$ absorbs the parasymplectic areas of the algebraic relations in $\pi_1(X)$. A choice of a rectifying cochain c for the surfacic coboundary yields a family of prequantum groupoids $\mathbf{T}_c$, each obtained as a quotient of the space of paths by a coherence relation $\sim_c$. We prove that each $\mathbf{T}_c$ is a fibrating diffeological groupoid, that its isotropy group is strictly connected and isomorphic to the total torus of periods $T_\omega = \mathbf{R}/P_\omega$, and that it carries a unique left- and right-invariant prequantum 1-form λ_c whose curvature is $\text{trg}^* \omega - \text{src}^* \omega$. The family $\{\mathbf{T}_c\}_c$ is an affine space directed by the character group $\text{Hom}(\pi_1(X), T_\omega)$.

The analysis of the symmetry group of these groupoids, and their comparison with the classical prequantum bundles of Kostant–Souriau, is deferred to a subsequent work.

INTRODUCTION

This paper is the continuation of [PIZ25a], in the framework of diffeology [PIZ13]. In Part I, we constructed the prequantum groupoid $\mathbf{T}_\omega$ for connected and simply connected parasymplectic spaces (X, ω) . The construction proceeds by distilling the space of paths $\text{Paths}(X)$, equipped with the Action Form $\mathbf{K}\omega$, into a diffeological groupoid whose space of morphisms is the quotient of $\text{Paths}(X)$ by the

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relation identifying paths with the same endpoints and the same action modulo periods.

The assumption of simple connectedness is restrictive. Many physically relevant systems, such as the torus or surfaces of higher genus, are not simply connected. In these cases, the geometry of the space X introduces new periods—surfacic periods—that are not captured by the spherical or toric cycles alone. A naive application of the Part I construction would fail to produce a connected isotropy group, and the quotient would not be a fibrating groupoid.

In this Part II, we extend the construction to arbitrary connected diffeological spaces. Our strategy relies on a direct path approach: the obstruction to the additivity of the prequantum action is identified directly on the space of loops, and measured by a cocycle τ on the fundamental group $\pi_1(X, x_0)$. By incorporating the values of τ on the group relations into the Total Group of Periods P_ω , we ensure that the prequantum structure closes on the quotient torus $T_\omega = \mathbf{R}/P_\omega$. A choice of rectifying cochain c for the surfacic coboundary then produces a prequantum groupoid $\mathbf{T}_c$, and varying c produces a family indexed by the character group $\text{Hom}(\pi_1(X), T_\omega)$.

The remainder of the paper is organized as follows. Section I constructs the three-tiered period hierarchy and the Total Group of Periods. Section II constructs the rectified integration function and the Chasles function Φ_c , whose vanishing defines the coherence relation. Section III defines the prequantum groupoids $\mathbf{T}_c$ and establishes their structural properties. Section IV applies the construction to the surface of genus 2 and to the torus T^2 . We close with a short conclusion.

We make no assumptions regarding the status of the objects as manifolds, infinite-dimensional manifolds, or Lie groups. All constructions are internal to the category of diffeological spaces, and the operations of taking quotients and forming functional spaces require no topological or regularity conditions to yield well-defined smooth structures.

I. THE THREE-TIERED PERIOD HIERARCHY

The torus of periods, quotient of the real line $\mathbf{R}$ by the group of periods P_ω , plays a fundamental role in geometric quantization as the universal carrier of the quantum phase. When X is simply connected, the periods originate entirely from smooth 2-spheres. In the general non-simply connected case, the periods undergo a canonical three-tiered filtration: spherical periods, toric periods, and surfacic periods.

We construct the total torus of periods through a sequence of successive diffeological subductions:

$$\mathbf{R} \twoheadrightarrow T_{\text{sph}} = \mathbf{R}/P_{\text{sph}} \twoheadrightarrow T_{\text{tor}} = \mathbf{R}/P_{\text{tor}} \twoheadrightarrow T_\omega = \mathbf{R}/P_\omega.$$

1. The Spherical Periods.

Let (X, ω) be a connected parasymplectic diffeological space. Recall the chain-homotopy operator $\mathbf{K}: \Omega^k(X) \rightarrow \Omega^{k-1}(\text{Paths}(X))$ and the Action Form $\mathbf{K}\omega \in \Omega^1(\text{Paths}(X))$.

We denote by $L_0 = \text{Loops}_0(X, x_0)$ the connected component of contractible loops based at a chosen basepoint $x_0 \in X$. The restriction of $\mathbf{K}\omega$ to L_0 is closed. The *Group of Spherical Periods* P_{sph} is the subgroup of $\mathbf{R}$ defined by integrating $\mathbf{K}\omega$ over all closed cycles in L_0 :

$$P_{\text{sph}} = \left\{ \int_{\sigma} \mathbf{K}\omega \mid \sigma \in \text{Loops}(L_0) \right\} \subset \mathbf{R}.$$

Geometrically, each cycle $\sigma \in \text{Loops}(L_0)$ represents a smooth 2-sphere in X , and $\int_{\sigma} \mathbf{K}\omega = \int_{S^2} \omega$. We assume that P_{sph} is a strict subgroup ($P_{\text{sph}} \neq \mathbf{R}$). The quotient $T_{\text{sph}} = \mathbf{R}/P_{\text{sph}}$ is the *Torus of Spherical Periods*. It carries a canonical closed 1-form $\theta_{\text{sph}} \in \Omega^1(T_{\text{sph}})$ characterized by $\pi_{\text{sph}}^*(\theta_{\text{sph}}) = dt$, where $\pi_{\text{sph}}: \mathbf{R} \rightarrow T_{\text{sph}}$ is the canonical projection.

2. The Toric Periods.

When $\pi_1(X) \neq 0$, the space of free loops decomposes into its connected components:

$$\text{Loops}(X) = \coprod_{i \in \pi_0(\text{Loops}(X))} L_i.$$

On each component L_i , the restriction of the closed 1-form $\mathbf{K}\omega$ defines a local period group:

$$P_i = \left\{ \int_{\sigma} \mathbf{K}\omega \mid \sigma \in \text{Loops}(L_i) \right\} \subset \mathbf{R}.$$

Geometrically, a cycle $\sigma \in \text{Loops}(L_i)$ corresponds to a smooth map from a 2-torus into X , $\Sigma: T^2 \rightarrow X$, and $\int_{\sigma} \mathbf{K}\omega = \int_{T^2} \Sigma^*(\omega)$.

The *Group of Toric Periods* P_{tor} is the subgroup of $\mathbf{R}$ generated by the union of all local periods:

$$P_{\text{tor}} = \left\langle \bigcup_{i \in \pi_0(\text{Loops}(X))} P_i \right\rangle.$$

Since the contractible component L_0 is one of these components, we have the canonical inclusion $P_{\text{sph}} \subset P_{\text{tor}} \subset \mathbf{R}$. We assume that P_{tor} is a strict subgroup. The quotient $T_{\text{tor}} = \mathbf{R}/P_{\text{tor}}$ is the *Torus of Toric Periods*. It is equipped with its canonical closed 1-form θ_{tor} characterized by $\pi_{\text{tor}}^*(\theta_{\text{tor}}) = dt$.

3. The Group of Indices.

To structure the homotopy of X , we fix a basepoint $x_0 \in X$ and consider the space of based loops $\text{Loops}(X, x_0)$. The set of connected components $\mathcal{J} = \pi_0(\text{Loops}(X, x_0))$ is canonically identified with the fundamental group $\pi_1(X, x_0)$. Concatenation of based loops induces a group multiplication on $\mathcal{J}$.

We choose a *basis of loops* $\mathcal{B} = \{\ell_i\}_{i \in \mathcal{J}}$ such that:

- (1) For each index $i \in \mathcal{I}$, the loop $\ell_i \in \text{Loops}(X, x_0)$ belongs to the connected component L_i .
- (2) For the identity element $0 \in \mathcal{I}$, we choose the constant loop $\ell_0 = x_0$.

4. The Intermediate Integration Function.

We define the intermediate integration function $\phi: \text{Loops}(X, x_0) \rightarrow T_{\text{tor}}$ component-wise relative to the chosen basis $\mathcal{B}$. For any loop ℓ in the component L_i (indexed by $i \in \mathcal{I}$):

$$\phi(\ell) = \pi_{\text{tor}} \left(\int_{\ell_i}^{\ell} \mathbf{K}\omega \right) \in T_{\text{tor}},$$

where $\pi_{\text{tor}}: \mathbf{R} \rightarrow T_{\text{tor}}$ is the canonical projection, and the integral is evaluated along any smooth path in the component L_i connecting the basis loop ℓ_i to ℓ .

Because any two paths connecting ℓ_i to ℓ within L_i differ by a closed cycle in $\text{Loops}(L_i)$, their integrals differ by an element of the local period group $P_i \subset P_{\text{tor}}$. Therefore, $\phi(\ell)$ is strictly well-defined in T_{tor} , independent of the path chosen within L_i . By construction, ϕ satisfies the differential identity $\phi^*(\theta_{\text{tor}}) = \mathbf{K}\omega \upharpoonright \text{Loops}(X, x_0)$.

5. The Surfacic Cocycle.

The function ϕ measures the parasymplectic action relative to the chosen basis $\mathcal{B}$. However, ϕ is generally not a homomorphism with respect to loop concatenation. Consider two based loops $\ell \in L_i$ and $\ell' \in L_j$. Their concatenation $\ell \vee \ell'$ belongs to the component $L_{i \cdot j}$. The integral defining $\phi(\ell \vee \ell')$ decomposes geometrically into two steps: first from the basis loop $\ell_{i \cdot j}$ to the concatenation $\ell_i \vee \ell_j$, and then from $\ell_i \vee \ell_j$ to $\ell \vee \ell'$. Using the additivity of path integration over concatenated homotopies established in Part I, the second term splits into the individual integrals modulo P_{tor} . Consequently, we obtain the fundamental relation:

$$\phi(\ell \vee \ell') = \phi(\ell) + \phi(\ell') + \tau(i, j),$$

where the defect of additivity is measured by the *surfacic cocycle* $\tau: \mathcal{I} \times \mathcal{I} \rightarrow T_{\text{tor}}$:

$$\tau(i, j) = \pi_{\text{tor}} \left(\int_{\ell_{i \cdot j}}^{\ell_i \vee \ell_j} \mathbf{K}\omega \right) \in T_{\text{tor}}.$$

Proposition (The Surfacic Cocycle). The map $\tau: \mathcal{I} \times \mathcal{I} \rightarrow T_{\text{tor}}$ is a group 2-cocycle of $\mathcal{I}$ with values in the abelian group T_{tor} . That is, for all $i, j, k \in \mathcal{I}$:

$$\tau(i, j) + \tau(i \cdot j, k) = \tau(j, k) + \tau(i, j \cdot k).$$

Note (Geometric Meaning). The cocycle τ admits a direct interpretation in terms of parasymplectic areas. If two elements $i, j \in \mathcal{I}$ commute, the difference $\tau(i, j) - \tau(j, i)$ is the area of the 2-torus swept out by sliding ℓ_i past ℓ_j . If i and j do not commute, the area is captured by the accumulated cocycle over a full relation in the fundamental group.

Proof. Let $i, j, k \in \mathcal{J}$. We evaluate the action of deforming $\ell_{i \cdot j \cdot k}$ to the triple concatenation $\ell_i \vee \ell_j \vee \ell_k$ along two distinct homotopic decomposition paths in $\text{Loops}(X, x_0)$.

1. *Left-parenthesized decomposition.* Decompose the deformation from $\ell_{(i \cdot j) \cdot k}$ to $(\ell_i \vee \ell_j) \vee \ell_k$ into two steps: first $\ell_{(i \cdot j) \cdot k} \rightarrow \ell_{i \cdot j} \vee \ell_k$, contributing $\tau(i \cdot j, k)$ modulo P_{tor} ; then $\ell_{i \cdot j} \vee \ell_k \rightarrow (\ell_i \vee \ell_j) \vee \ell_k$, contributing $\tau(i, j)$ by additivity. The total is $\tau(i, j) + \tau(i \cdot j, k)$.

2. *Right-parenthesized decomposition.* Similarly, the deformation from $\ell_{i \cdot (j \cdot k)}$ to $\ell_i \vee (\ell_j \vee \ell_k)$ through $\ell_i \vee \ell_{j \cdot k}$ contributes $\tau(i, j \cdot k) + \tau(j, k)$.

3. *Associativity homotopy.* The two endpoints $(\ell_i \vee \ell_j) \vee \ell_k$ and $\ell_i \vee (\ell_j \vee \ell_k)$ are connected by the canonical smooth reparametrization $\alpha: [0, 1] \rightarrow \text{Loops}(X, x_0)$. The associated 2-plot $\Phi(s, t) = \alpha(s)(t)$ factors through a 1-plot, so its pullback of ω vanishes identically. Hence $\int_{\alpha} \mathbf{K}\omega = 0$.

4. *Conclusion.* The closed cycle formed by the first path, the associativity homotopy, and the reverse of the second path lies in $L_{i \cdot j \cdot k}$, so its total integral is zero modulo P_{tor} . Hence $\tau(i, j) + \tau(i \cdot j, k) = \tau(j, k) + \tau(i, j \cdot k)$ in T_{tor} . $\square$

6. The Central Extension.

The 2-cocycle τ defines a group extension $\tilde{\Gamma}$ of the fundamental group $\mathcal{J}$ by the torus of toric periods T_{tor} . As a set, $\tilde{\Gamma} = \mathcal{J} \times T_{\text{tor}}$, equipped with the multiplication law:

$$(i, u) \cdot (j, v) = (i \cdot j, u + v + \tau(i, j)).$$

This extension is strictly central: the fiber elements $(0, a)$ commute with all elements of $\tilde{\Gamma}$.

7. The Total Group of Periods.

Let $F(\mathcal{J})$ be the free group generated by $\mathcal{J}$, and let $\mathcal{R} \subset F(\mathcal{J})$ be the normal subgroup of relations, so that $\mathcal{J} \simeq F(\mathcal{J})/\mathcal{R}$. Let $\sigma: \mathcal{J} \rightarrow \tilde{\Gamma}$ be the canonical section $\sigma(i) = (i, 0)$. By the universal property of free groups, σ extends uniquely to a group homomorphism $\Psi: F(\mathcal{J}) \rightarrow \tilde{\Gamma}$. Because Ψ lifts the canonical projection, it maps $\mathcal{R}$ into the central fiber T_{tor} .

For $w \in \mathcal{R}$, the *accumulated cocycle* $\mathcal{T}(w) \in T_{\text{tor}}$ is defined by $\Psi(w) = (0, \mathcal{T}(w))$.

Definition Total Group of Periods. Let $H_{\text{surf}} \subset T_{\text{tor}}$ be the subgroup generated by the accumulated cocycle values on all relations:

$$H_{\text{surf}} = \langle \{\mathcal{T}(w) \mid w \in \mathcal{R}\} \rangle.$$

The *Total Group of Periods* is the preimage of H_{surf} in $\mathbf{R}$ under π_{tor} :

$$P_{\omega} = \pi_{\text{tor}}^{-1}(H_{\text{surf}}) \subset \mathbf{R}.$$

We assume $P_\omega \neq \mathbf{R}$. The quotient $T_\omega = \mathbf{R}/P_\omega \simeq T_{\text{tor}}/H_{\text{surf}}$ is the *Total Torus of Periods*.

Note (Algebraic Significance). By construction, the projection of τ to T_ω is a coboundary: $[\pi_\omega \circ \tau] = 0 \in H^2(\pi_1(X), T_\omega)$. Equivalently, there exists a 1-cochain $c: \mathcal{J} \rightarrow T_\omega$ such that:

$$\pi_\omega(\tau(i, j)) = c(i) + c(j) - c(i \cdot j).$$

This cochain is the datum that rectifies the integration function, as explained in Section II.

II. RECTIFIED INTEGRATION AND CHASLES FUNCTION

With the Total Torus of Periods T_ω established, we construct the rectified integration function on the based loop space. Because the surfacic cocycle τ becomes a coboundary when projected to T_ω , the naive integration can be rectified into a genuine homomorphism with respect to loop concatenation. The rectified function is then extended to free loops, and used to define the Chasles function on pairs of paths with common endpoints.

8. The Rectified Integration on Based Loops.

By the construction of the Total Group of Periods P_ω , the projection of the surfacic cocycle to T_ω is a coboundary. Therefore, there exists a 1-cochain $c: \mathcal{J} \rightarrow T_\omega$ such that:

$$\pi_\omega(\tau(i, j)) = c(i) + c(j) - c(i \cdot j),$$

for all $i, j \in \mathcal{J}$. We fix such a cochain once and for all, normalized by $c(0) = 0$ for the identity element.

Recall the integration function $\phi: \text{Loops}(X, x_0) \rightarrow T_{\text{tor}}$, satisfying the identity $\phi(\ell \vee \ell') = \phi(\ell) + \phi(\ell') + \tau(i, j)$ for $\ell \in L_i$ and $\ell' \in L_j$. Project this function on the Total Torus of Periods and let $\phi' = \pi_\omega \circ \phi$. Then:

$$\begin{aligned} \phi'(\ell \vee \ell') &= \phi'(\ell) + \phi'(\ell') + \pi_\omega(\tau(i, j)) \\ &= \phi'(\ell) + \phi'(\ell') + c(i) + c(j) - c(i \cdot j). \end{aligned}$$

That is:

$$\phi'(\ell \vee \ell') + c(i \cdot j) = (\phi'(\ell) + c(i)) + (\phi'(\ell') + c(j)).$$

Defining then

$$\phi_c(\ell) = \phi'(\ell) + c(i) \quad \text{for all } \ell \in L_i,$$

we obtain the following proposition.

Proposition (Properties of ϕ_c). The map ϕ_c satisfies:

- (1) **Normalization:** $\phi_c(x_0) = 0$.
- (2) **Additivity:** for all $\ell, \ell' \in \text{Loops}(X, x_0)$, $\phi_c(\ell \vee \ell') = \phi_c(\ell) + \phi_c(\ell')$.

- (3) **Primitive property:** $\phi_c^*(\theta_\omega) = \mathbf{K}\omega \upharpoonright \text{Loops}(X, x_0)$, where θ_ω is the canonical 1-form on T_ω .
- (4) **Family:** If c' is another cochain satisfying the same coboundary relation, then $\chi = c' - c$ is a character $\mathcal{J} \rightarrow T_\omega$, and $\phi_{c'} = \phi_c + \chi \circ \text{comp}$, where $\text{comp}: \text{Loops}(X, x_0) \rightarrow \mathcal{J}$ is the component map. The family $\{\phi_c\}_c$ is an affine space directed by $\text{Hom}(\mathcal{J}, T_\omega)$.

Proof. (1) *Normalization.* The constant loop $\mathbf{x}_0$ belongs to L_0 , and $c(0) = 0$, so $\phi_c(\mathbf{x}_0) = 0$.

(2) *Additivity.* Immediate from the identity established above.

(3) *Primitive property.* On each component L_i , the cochain $c(i)$ is constant, so its differential vanishes. Hence $\phi_c^*(\theta_\omega) \upharpoonright L_i = \phi'^*(\theta_\omega) \upharpoonright L_i = \phi^*(\theta_{\text{tor}}) \upharpoonright L_i = \mathbf{K}\omega \upharpoonright L_i$. Summing over components gives the result.

(4) *Family.* If c' satisfies the same coboundary relation, then $\chi = c' - c$ satisfies $\chi(i) + \chi(j) - \chi(i \cdot j) = 0$, so χ is a homomorphism. The relation $\phi_{c'} = \phi_c + \chi \circ \text{comp}$ is immediate from the definition. Freeness and transitivity of the action follow. $\square$

9. Extension to the Free Loop Space.

The rectified integration ϕ_c is defined on the based loop space. We extend it to the free loop space $\text{Loops}(X)$.

Let $\ell \in \text{Loops}(X)$, and let $x = \ell(0) = \ell(1)$ be its basepoint. Choose a smooth path $\delta \in \text{Paths}(X)$ from x_0 to x . The concatenation $\delta \vee \ell \vee \bar{\delta}$ is a loop based at x_0 . We define the value of the free loop integration function by:

$$\phi_c(\ell) = \phi_c(\delta \vee \ell \vee \bar{\delta}) \in T_\omega.$$

Proposition (Independence of the connecting path). The extension is well-defined and independent of the choice of δ . For this reason we denote it by the same symbol ϕ_c .

Proof. Let δ' be another path from x_0 to x . The two loops $\delta' \vee \ell \vee \bar{\delta}'$ and $C \vee (\delta \vee \ell \vee \bar{\delta}) \vee \bar{C}$, where $C = \delta' \vee \bar{\delta}$, differ by the insertion of a backtrack. The integral of $\mathbf{K}\omega$ along a backtrack vanishes, because it amounts to integrating the 2-form ω on a 1-plot. Since ϕ_c is additive and T_ω is abelian, the two values coincide. The extension is unique: it is the only map $\text{Loops}(X) \rightarrow T_\omega$ whose restriction to $\text{Loops}(X, x_0)$ is ϕ_c and which is additive under concatenation. $\square$

10. The Chasles Function.

We recall the space of pairs of paths with coincident endpoints:

$$\text{Paths}_2(X) = \{(\gamma, \gamma') \in \text{Paths}(X) \times \text{Paths}(X) \mid \text{ends}(\gamma) = \text{ends}(\gamma')\}.$$

Let $(\gamma, \gamma') \in \text{Paths}_2(X)$, so that $\gamma(0) = \gamma'(0) = x$ and $\gamma(1) = \gamma'(1) = x'$. The concatenation $\gamma \vee \tilde{\gamma}'$ forms a closed loop based at x .

We define the *Chasles function* $\Phi_c: \text{Paths}_2(X) \rightarrow T_\omega$ by:

$$\Phi_c(\gamma, \gamma') = \phi_c(\gamma \vee \tilde{\gamma}'),$$

where ϕ_c is the free loop integration function of Article 9. The definition involves no auxiliary path, because ϕ_c is defined directly on the free loop space.

Proposition (Chasles Identity). For any three paths $\gamma, \gamma', \gamma''$ with the same endpoints:

$$\Phi_c(\gamma, \gamma') + \Phi_c(\gamma', \gamma'') = \Phi_c(\gamma, \gamma'').$$

Remark (Family). If c' is another cochain and $\chi = c' - c$, then $\Phi_{c'}(\gamma, \gamma') = \Phi_c(\gamma, \gamma') + \chi(\gamma \vee \tilde{\gamma}')$. The family $\{\Phi_c\}_c$ is an affine space directed by $\text{Hom}(\mathcal{J}, T_\omega)$.

Proof. The loop $\gamma \vee \tilde{\gamma}''$ and the concatenation $(\gamma \vee \tilde{\gamma}') \vee (\gamma' \vee \tilde{\gamma}'')$ differ by a backtrack, whose integral vanishes. By additivity of ϕ_c on free loops sharing a basepoint, $\Phi_c(\gamma, \gamma'') = \Phi_c(\gamma, \gamma') + \Phi_c(\gamma', \gamma'')$. $\square$

III. THE PREQUANTUM GROUPOIDS

All the geometric ingredients are now in place to construct the prequantum groupoids for an arbitrary connected parasymplectic space (X, ω) . We distill the Primordial Quantum Broth $(\text{Paths}(X), \mathbf{K}\omega)$ into a family of prequantum groupoids $\{\mathbf{T}_c\}_c$, one for each choice of cochain c resolving the surfacic coboundary.

11. The Quantum Broth.

Before crystallizing the structure into a groupoid, we recognize the fundamental object from which all quantum geometry derives. We call the pair $(\text{Paths}(X), \mathbf{K}\omega)$ the *Primordial Quantum Broth*.¹ This infinite-dimensional space contains every possible transition of the system. By integration, the Action Form $\mathbf{K}\omega$ assigns a potential phase to every finite fluctuation of a motion. However, this space is noisy: it contains infinitely many distinct paths that are physically indistinguishable because they differ by variations carrying no parasymplectic action modulo periods.

The construction of the prequantum groupoids is the process of *distilling* this broth. We identify paths that carry the same phase information, collapsing the infinite redundancy of the broth into the precise geometry of the groupoid.

¹This structure has a concrete existence in diffeology; it is not merely a figure of speech. Diffeology was expressly developed to handle such infinite-dimensional structures—a functional space equipped with a genuine differential form—with simplicity and rigor.

12. The Equivalence Relation.

For each cochain c resolving the surfacic coboundary, we define the *coherence relation* $\sim_c$ on $\text{Paths}(X)$:

$$\gamma \sim_c \gamma' \text{ if and only if } \text{ends}(\gamma) = \text{ends}(\gamma') \text{ and } \Phi_c(\gamma, \gamma') = 0 \in T_\omega.$$

Here, $\text{ends}(\gamma) = (\gamma(0), \gamma(1)) \in X \times X$ is the ends map, and $\Phi_c: \text{Paths}_2(X) \rightarrow T_\omega$ is the Chasles function of Article 10.

This relation identifies paths sharing the same endpoints whose relative parasymplectic action, corrected by the cochain c on the components of the loop space, vanishes in the Total Torus of Periods.

13. Definition of the Groupoids.

Given a resolution c of the surfacic cocycle τ , the quotient of the space of paths by the coherence relation $\sim_c$ defines the morphisms of the prequantum groupoid.

Definition (The Prequantum Groupoids). The *Prequantum Groupoid* $\mathbf{T}_c$ is the category internal to diffeology defined by:

$$\text{Obj}(\mathbf{T}_c) = X, \quad \text{Mor}(\mathbf{T}_c) = \mathcal{Y}_c = \text{Paths}(X) / \sim_c.$$

The source and target maps $\text{src}, \text{trg}: \mathcal{Y}_c \rightarrow X$ are induced by the end-points evaluation maps $\hat{0}, \hat{1}: \text{Paths}(X) \rightarrow X$:

$$\text{src}([\gamma]_c) = \gamma(0), \quad \text{trg}([\gamma]_c) = \gamma(1).$$

The composition of composable morphisms is induced by the concatenation of paths:

$$[\gamma]_c \cdot [\gamma']_c = [\gamma \vee \gamma']_c, \quad \text{for } \gamma(1) = \gamma'(0).$$

The unit map $x \mapsto \mathbf{1}_x = [x]_c$ assigns to each classical motion $x \in X$ the class of the constant path at x .

14. Structural Properties.

The resulting object is a fibrating diffeological groupoid equipped with a prequantum 1-form.

Theorem (The Prequantum Structure). Let (X, ω) be a connected parasymplectic diffeological space with $\omega \neq 0$, and let c be a resolution of the surfacic coboundary. The groupoid $\mathbf{T}_c$ satisfies:

- (1) It is a fibrating diffeological groupoid over X : the projection $\text{ends}: \mathcal{Y}_c \rightarrow X \times X$ is a subduction.
- (2) The isotropy group at any point $x \in X$, $\mathbf{T}_{c,x}$, is strictly connected and isomorphic to the Total Torus of Periods T_ω via the descent of ϕ_c .
- (3) There exists a unique left- and right-invariant 1-form $\lambda_c \in \Omega^1(\mathcal{Y}_c)$, the *prequantum 1-form*, such that $\text{class}_c^*(\lambda_c) = \mathbf{K}\omega$, where $\text{class}_c: \text{Paths}(X) \rightarrow \mathcal{Y}_c$ is the quotient projection.

(4) The curvature of λ_c satisfies $d\lambda_c = \text{trg}^* \omega - \text{src}^* \omega$.

Proof. 1. *Groupoid structure and subduction.* The additivity of ϕ_c guarantees that $\sim_c$ is compatible with concatenation. The ends map factors through the quotient, and since $\text{ends}: \text{Paths}(X) \rightarrow X \times X$ is a subduction, its factorization on $\mathcal{U}_c$ is also a subduction.

2. *Isotropy.* The isotropy at x_0 consists of the morphism classes of loops in $\text{Loops}(X, x_0)$. Two classes coincide iff $\Phi_c(\ell, \ell') = 0$, which by additivity of ϕ_c is equivalent to $\phi_c(\ell) = \phi_c(\ell')$. Hence the induced map $\overline{\phi_c}: \mathbf{T}_{c, x_0} \rightarrow \text{im}(\phi_c)$ is well-defined and injective. That $\text{im}(\phi_c) = T_\omega$ follows from the same argument as in the simply connected case, using the contractible component L_0 and the fact that the path-primitive $\mathbf{F}$ on L_0 has image all of $\mathbf{R}$. The extension to arbitrary x uses conjugation by a path from x to x_0 , which preserves $\sim_c$ because Φ_c is conjugation-invariant.

3. *The prequantum form.* Apply the diffeological descent criterion to $\mathbf{K}\omega$ and the subduction class_c . If $\text{class}_c \circ P = \text{class}_c \circ P'$ for two plots P, P' , then for all r , $\text{ends}(P(r)) = \text{ends}(P'(r))$ and $\Phi_c(P(r), P'(r)) = 0$. Hence $\int_{P'(r)}^{P(r)} \mathbf{K}\omega \in P_\omega$, which is locally constant because P_ω is discrete. The exterior derivative vanishes, so $\mathbf{K}\omega(P) = \mathbf{K}\omega(P')$ on each connected domain, and the descent criterion applies.

4. *Curvature.* By the chain-homotopy identity, $d\mathbf{K}\omega = \hat{1}^* \omega - \hat{0}^* \omega$. Pulling back along class_c and using the relation $\hat{1}_{\text{Paths}} = \text{trg} \circ \text{class}_c$, $\hat{0}_{\text{Paths}} = \text{src} \circ \text{class}_c$, we get $\text{class}_c^*(d\lambda_c) = \text{class}_c^*(\text{trg}^* \omega - \text{src}^* \omega)$. Since class_c is a subduction, $d\lambda_c = \text{trg}^* \omega - \text{src}^* \omega$. $\square$

Remark (The family). The construction yields one prequantum groupoid for each cochain c resolving the surfacic coboundary. The family $\{\mathbf{T}_c\}_c$ is parametrized by the character group $\text{Hom}(\pi_1(X), T_\omega)$ once a base cochain is fixed, and the whole family is an affine space directed by that group. The study of the symmetry group of these groupoids, the comparison with the classical prequantum bundles of Kostant and Souriau, and the analysis of the moduli space of the family, are the subject of a subsequent work.

IV. EXAMPLES

We illustrate the construction with two archetypal examples: a surface of genus 2, where the quantum phase is generated purely by the surfacic periods, and the torus T^2 , where the three tiers collapse into one.

15. Explicit Construction: The Surface of Genus 2.

Let Σ_2 be a compact, orientable surface of genus 2, equipped with an area form ω . This example illustrates the mechanics of the Total Group of Periods, because the

spherical and toric periods vanish completely, leaving only the surfacic periods to generate the quantum phase.

1. Vanishing of spherical and toric periods. The surface Σ_2 has the hyperbolic plane $\mathbf{H}^2$ as its contractible universal cover, so $\pi_2(\Sigma_2) = 0$ and $P_{\text{sph}} = \{0\}$. Every smooth map $T^2 \rightarrow \Sigma_2$ has degree zero (because every abelian subgroup of a surface group of genus at least two is cyclic), so $P_{\text{tor}} = \{0\}$ and $T_{\text{tor}} = \mathbf{R}$.

2. Surfacic period. The fundamental group $\pi_1(\Sigma_2, x_0)$ is generated by four loops a_1, b_1, a_2, b_2 with the single defining relation $R = [a_1, b_1] \cdot [a_2, b_2] = 0$. Geometrically, the relation corresponds to the concatenated loop γ_R , whose lift to $\mathbf{H}^2$ is the boundary of the fundamental octagon. The accumulated cocycle is:

$$\mathcal{T}(R) = \int_{x_0}^{\gamma_R} \mathbf{K}\omega = \iint_{\text{Octagon}} \tilde{\omega} = \text{Area}(\Sigma_2) = A.$$

Hence $P_\omega = A\mathbf{Z}$ and $T_\omega = \mathbf{R}/A\mathbf{Z}$.

3. Contractible loops sweep out $\mathbf{R}$. A loop $\ell \in L_0$ lifts to a closed loop $\tilde{\ell}$ in $\mathbf{H}^2$ bounding a disk D . Since $\mathbf{H}^2$ has infinite area, there is a smooth 1-parameter family of loops whose bounded areas cover all of $\mathbf{R}$. Projecting back to Σ_2 and to T_ω , the image of L_0 under ϕ_c covers the whole torus.

4. Restoring the fundamental group relation. The concatenation of the eight generator loops yields γ_R , whose class must equal the identity morphism $\mathbf{1}_{x_0}$ for the groupoid to faithfully represent $\pi_1(\Sigma_2)$. The phase difference between γ_R and the constant loop is A . Quotienting by $P_\omega = A\mathbf{Z}$ forces $\phi_c(\gamma_R) \equiv 0$, identifying the octagon boundary with the constant loop.

16. The Torus T^2 .

Let $X = \mathbf{R}^2/\mathbf{Z}^2$ be the standard torus equipped with the symplectic area form $\omega = dx \wedge dy$. The fundamental group is $\pi_1(X) \simeq \mathbf{Z}^2$, generated by the two canonical loops A and B . The toric periods are $P_{\text{tor}} = \mathbf{Z}$.

The fundamental relation of the torus is the commutator $ABA^{-1}B^{-1} = 1$, and the surfacic period associated with it is the area of the fundamental domain, which is 1. Since $1 \in \mathbf{Z} = P_{\text{tor}}$, the surfacic contribution does not enlarge the period group. Therefore:

$$P_\omega = P_{\text{tor}} = \mathbf{Z}.$$

The prequantum isotropy is $T_{c,x} \simeq \mathbf{R}/\mathbf{Z} \simeq S^1$, recovering the standard prequantization of the torus. This contrasts with higher genus surfaces, where the surfacic period is strictly larger than the toric period.

CONCLUSION

This paper has constructed the prequantum groupoids T_c for arbitrary connected parasymplectic diffeological spaces. The construction proceeds by disarticulating

the periods of the parasymplectic form into a three-tiered hierarchy (spherical, toric, and surfacic), incorporating the surfacic cocycle into the Total Group of Periods, rectifying the naive integration function by a choice of cochain c , and quotienting the space of paths by the resulting coherence relation. Each resulting groupoid is fibrating, its isotropy group is the total torus of periods, and it carries a canonical prequantum 1-form whose curvature is the difference of the parasymplectic form on the target and source.

The family of prequantum groupoids is indexed by the character group of the fundamental group, and the whole family forms an affine space directed by this group. The analysis of the symmetry group of these groupoids, the identification with the classical prequantum bundles of Kostant and Souriau, and the study of the moduli space of the family, will be presented in subsequent work.

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